

SHEREEN ANAND

PES University, Bangalore

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Research Interests

Artificial Intelligence, Machine Learning, Large Language Models, Multimodal Learning, Reinforcement Learning, Natural Language Processing, Computer Vision, Reliable and Scalable AI Systems, Applied Deep Learning, and AI for Real-World.

Education

Bachelor of Technology in Computer Science and Engineering

PES University, Bangalore

Sept 2022 – Jul 2026

CGPA: 3.56/5.0

KSEAB (PUC)

RV PU College

June 2020 - June 2022

Score: 95.3%

CBSE (Class 10)

Sri Kumaran's Children's Home

May 2018 - May 2020

Score: 93%

Experience

Researcher

Microsoft Research (Manager: Gopal Srinivasan)

Sept 2026 – Present

Bangalore, India

- Working on the Copilot coding agent to enhance AI-assisted software development and developer productivity.

Applied Scientist Intern, Central Machine Learning Team

Amazon (Manager: Kaveri Sivaramakrishnan)

Jan 2026 – Jun 2026

Bangalore, India

- Developed ASPEX**, an aspect-grounded persona generation framework integrating hierarchical taxonomy induction with soft assignment, and prompt optimization OPRO technique to model multi-faceted user preferences for personalized conversational recommendation.
- Designed and benchmarked persona-conditioned recommendation across **8 downstream personalization tasks** spanning **Amazon and 3 public datasets** with a cumulative of **300K+ users**, enforcing strict temporal separation between persona construction and downstream evaluation.
- Established state-of-the-art performance over contemporary persona generation baselines, achieving relative **24.2%** PR-AUC and **15.8%** ROC-AUC improvement in the engagement prediction. Observed **18.9%** higher recommendation follow-up rate in suggestions provided.
- Optimized production LLM inference on AWS Bedrock using Claude models, reducing end-to-end latency by 20.3% using an aspect-grounded verification methodology; work submitted to the **EACL Industry Track 2027**.

Student Researcher, Autonomy and Quality for Agents

Google DeepMind (Manager: Sagar Gubbi)

Jun 2025 – Dec 2025

Bangalore, India

- Architected VISTA-Bench**, an adversarial evaluation framework for frontier agentic AI that auto-generates distributionally diverse testing environments using a dynamic Symmetric Adversarial Protocol.
- Engineered scalable evaluation pipelines across **300+ synthetic mobile environments**; formalized automated telemetry metrics for high-dimensional compositional generalization, behavioral trajectory consistency, and multi-step error recovery dynamics, minimizing empirical evaluation overhead by **35%**.
- Identified core brittleness modes in state-of-the-art frontier agents, systematically characterizing performance degradation profiles to prove that despite achieving **90%+ standard benchmark accuracy**, agent policy success rates decayed by up to **42%** under non-stationary covariate distribution shifts.
- Discovered and quantified a multilingual robustness paradox, establishing a statistically rigorous counter-intuitive phenomenon where frontier agents manifest a **22% increase** in out-of-distribution generalization vectors when evaluated using low-resource languages compared to high-resource baselines.

Research Experience

Research Intern

Advisor: Prof. V. Ramasubramanian, Summer Research Internship Program

IIIT Bangalore

Bangalore, India

- Constructed low-resource multilingual ASR systems leveraging **matching networks** and **self-supervised learning** to address data scarcity in underserved linguistic communities.
- Engineered cross-lingual transfer mechanisms using **wav2vec 2.0** with alignment-aware modeling, reducing **phoneme error rates by 28%** in complex code-switching scenarios.
- Analyzed phonetic confusion patterns across **15 language pairs**, identifying systematic misalignments in acoustic feature representations that propagate under extreme data scarcity (**<100 hours** labeled audio).
- Demonstrated that **self-supervised pretraining** can democratize speech technology access for **11 low-resource languages**, establishing baselines for future language-specific acoustic modeling.

Publications

Anand, S., Vamshi, V., Peddi, V., Prabhakaran, V. (2025). Enhanced Trading Systems with Intelligent Market Analysis. In: *Proceedings of the 4th International Conference on Artificial Intelligence and Applications (ICAIA)*. **Accepted (Main Track)**.

Anand, S., Mourya, A., Mukerjee, R., Puranik, V., Kaveri, S. (2027). ASPEX: Aspect-Grounded Persona Generation with Hierarchical Taxonomy. In: *EACL Industry Track 2027*. **Submitted**. [OpenReview]

Projects

Deforestation Detection & Forecasting from Satellite Imagery | *Python, TensorFlow, GDAL, U-Net* **2024**

- Aggregated and preprocessed multi-temporal Sentinel-2 and Landsat imagery across **12 districts**, applying cloud-masking and NDVI-based filtering to construct a clean, geo-referenced training corpus.
- Trained a **U-Net segmentation model** to delineate forest cover loss at pixel level, reaching **89% IoU** on held-out regions through iterative tuning of encoder depth and loss weighting for class imbalance.
- Formulated an LSTM-based forecasting layer atop segmentation outputs to project deforestation trends 6 months ahead, cutting mean prediction error by **21%** against a linear-regression trend baseline evaluated on the same held-out districts.

Skin Cancer Detection using Deep Learning | *Python, PyTorch, CNN* **2023**

- Curated and preprocessed a dermoscopic image dataset of **10K+ lesion samples**, applying targeted augmentation (rotation, color-jitter, oversampling) to correct class imbalance in minority melanoma cases.
- Fine-tuned a **CNN classifier** (EfficientNet backbone) via transfer learning to distinguish malignant from benign lesions, attaining **92% classification accuracy** and **0.90 F1-score** on a stratified test split.
- Applied **Grad-CAM-based interpretability** to visualize activation regions driving predictions, cross-validating model attention against dermatologist-annotated lesion boundaries to check for spurious correlation.

Teaching & Mentorship

Student Mentor **IEEE RAS**

PES University *2024-2025*

- Guided junior students through robotics competition projects and provided technical mentorship. Conducted workshops on robot programming.

Teaching Assistant **Practical Applied Machine Learning**

Centre for Entrepreneurship and Innovation, PES University *2024-2025*

- Designed lab exercises and practical sessions for 55+ students covering ML concepts. Led hands-on tutorials implementing ML algorithms provided one-on-one guidance on project development and debugging.

Technical Skills

Programming Languages: Python, C++, R, MATLAB, SQL, Java, JavaScript, TypeScript, Go, Rust, Scala, HTML/CSS

Machine Learning & AI: TensorFlow, PyTorch, scikit-learn, Keras, XGBoost, LightGBM, reinforcement learning, neural networks, ensemble methods, feature selection, hyperparameter optimization

Data Engineering: Apache Spark, Hadoop, Airflow, Pandas, NumPy, SciPy, real-time streaming, event processing, data quality checks, feature stores

Visualization & Analytics: Matplotlib, Seaborn, Plotly, Tableau, Power BI, interactive dashboards

Backend Development: Python, C++, Java, microservices, REST APIs, GraphQL, message queues

Cloud & Infrastructure: AWS (EC2, S3, Lambda, RDS, SQS), GCP (BigQuery, Dataflow), Azure, Docker, Kubernetes, Terraform, real-time data pipelines

Databases: PostgreSQL, MySQL, MongoDB, Redis, Cassandra, InfluxDB, KDB+/q, time-series databases

System Design: High-frequency trading infrastructure, low-latency systems, distributed computing, microservices, load balancing, caching, fault tolerance

Development Tools: Git, GitHub, VSCode, IntelliJ, Jupyter, Docker, CI/CD, testing frameworks (Pytest, Jest), profiling

Operating Systems: Linux, Windows, macOS, Shell scripting, system administration

Achievements & Awards

- Runner-Up – Amazon Internal Hackathon 2026
- Top-100 – Google Girl Hackathon 2025 Software Round
- Finalist – Smart India Hackathon in Software Edition 2023, selected at national level.
- Runner-Up – Robowars Tournament in 2023
- Finalist – Srinivasa Ramanujan Mathematics Competition (SRMC) 2016, invited to offline finals at IIT Bombay.